International Journal of Theoretical and Applied Research (IJTAR)
ISSN: 2812-5878

Homepage: https://ijtar.journals.ekb.eg

Original article

An Intelligent Budget Management Mobile Application Based on a Recurrent Neural Network

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ARTICLE INFO

Received 02/10/2025
Revised 16/11/2025
Accepted 20/11/2025

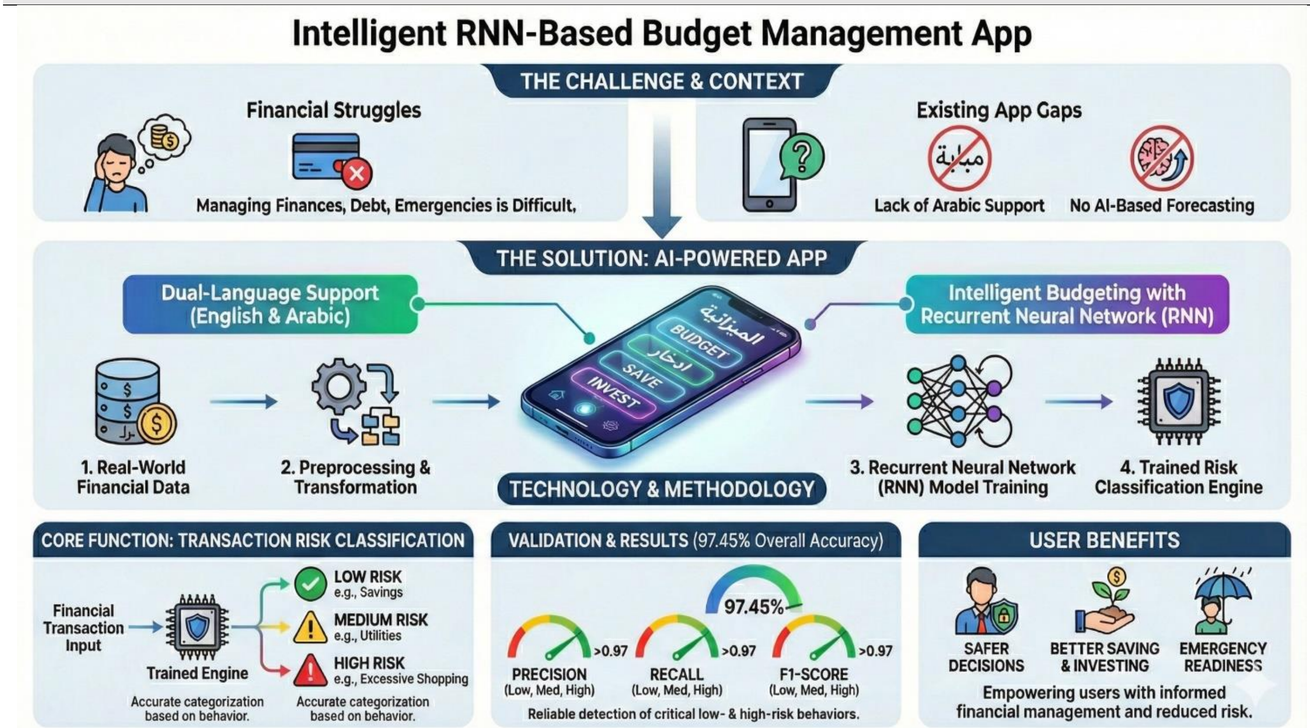
Keywords

Budget managemen
Mobile application
Artificial intelligence
Recurrent neural network

ABSTRACT

Budgeting is important for both people and businesses to manage their finances, save, stay out of debt, and be ready for financial emergencies. This paper aims to propose an intelligent mobile application to reduce financial risks and support better ways of saving and investing through intelligent budgeting. A lot of budgeting applications lacked Arabic language support and AI-based forecasting capabilities. This gap is addressed by the development of an application that supports English and Arabic speakers through a Recurrent Neural Network (RNN) for enhanced financial and management capabilities. The model was trained on a real-world financial dataset, preprocessing it, and transforming its data. The results demonstrated that the model was highly accurate in the classification of financial transactions based on the level of risk. The application was tested on new data; it reached an overall accuracy of 97.45%. Precision, recall, and F1 measures are all higher than 0.97 for each of the risk categories: low, medium, and high. These results validate the reliability of the system, particularly in detecting critical low- and high-risk behaviors, and its capacity to help users make safer and more informed decisions regarding their finances.

Graphical abstract



1. Introduction

Budgeting skills should be applied to individuals and families so that everyone can manage their finances effectively. These skills demand much discipline and self-awareness, and without these skills, the person may face some financial difficulties, which in the long run may influence the quality of life and achievement of their dreams. The economic challenges today are very different, and

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DOI: 10.21608/IJTAR.2025.427658.1148

needs are growing in the modern world. People require smart and effective ways and means to manage the available resources to provide a secure and strong financial foundation [1]. The reality of life is that living has become costly, and the economic stresses call for people to learn how to manage their scarce resources. All budgeting skills are no longer considered luxuries but a necessity for living a comfortable and secure future life. Personal attributes such as financial literacy, mental budgeting, and self-control are good in that they create a link between achievements in the workplace. A study suggests that people with high financial literacy have better financial health than others. In other words, understanding the basic aspects of finances, personal planning, saving, and investing can help enhance one's finances [2]. Budgeting has undergone significant changes over the years with the advancement of technology. Traditionally, budgeting began with a hierarchical process based on top-down instructions, which made it difficult to adapt to changes. Technological advancements have brought about many changes in budget management, with many software solutions being introduced to plan, predict financial crises, and evaluate user spending [3].

This paper proposes a budget management mobile application based on a Recurrent Neural Network (RNN), which belongs to a family of neural architectures specifically designed for sequential and temporal data processing. Despite their usefulness in areas such as RNNs are characterized by recurrent connections that enable information from previous time steps to influence current outputs, allowing the model to capture contextual and time-dependent patterns effectively [4]. Although RNNs have proven effective in domains such as natural language processing, speech modelling, and time-series forecasting, they often face training challenges, particularly vanishing and exploding gradient problems [5, 6]. To address these limitations, advanced architectures such as Long Short-Term Memory (LSTM) [7] and Gated Recurrent Units (GRU) [8] have been developed. These architectures enhance the network's ability to learn and retain long-term dependencies, often outperforming traditional RNNs in practical applications [6]. The proposed mobile application enables users to manage their finances intelligently through a user-friendly and intuitive interface. It provides multiple services, including recording expenditures across categories such as necessities, loans, and discretionary spending within a monthly adjustable budget. The application tracks both daily and monthly expenditures across all categories and generates detailed reports that help users monitor their financial status and spending patterns. One of the most important tools provided in the application is the opportunity to analyze spending data and even foresee the potential crises that might occur due to spending and changes in income. Thus, the proposed Budget Management Application serves as a comprehensive, user-oriented tool designed to help individuals enhance their budgeting skills and financial awareness. By integrating intelligent, technology-driven solutions, the system empowers users to build long-term financial literacy and make informed, sustainable financial decisions. These features make the application a reliable and effective partner in achieving financial stability and long-term economic well-being.

Many applications can help with money management, but there is no comprehensive application that provides several important features. It's crucial to develop a comprehensive application that is equipped with the latest technologies and offers various languages to engage with various segments. For example, most of them are not in Arabic. This requires the urgency of developing an Arabic-language application with extensive customization options to accommodate various cultures and lifestyles. Although there are a few applications that use AI to predict financial crises or their imminence, they do not offer personal plans and purchase recommendations that fit the user's budget, nor do they provide the user with an alert when the transaction exceeds the budget, nor give a retirement financial plan. Thus, this is a very serious gap as it is challenging for users to reach their financial objectives in a way that works for them. This suggests that these features were not given enough consideration when earlier applications were developed. Therefore, this project proposes a solution by developing a new application that combines the features from earlier versions with several fundamental features in each application to give users the best possible assistance in managing their financial affairs and reaching their financial objectives. The remainder of this paper is structured as follows: Section 2 presents many different budget management applications and papers. Section 3 proposes the architecture of the proposed application. In Section 4, we present the experimental results and discussion of the proposed application. Finally, Section 5 provides a conclusion.

2. Related works

In this section, the literature review will be presented. It is based on two main categories: Research papers and application papers, as shown in Fig. 1.

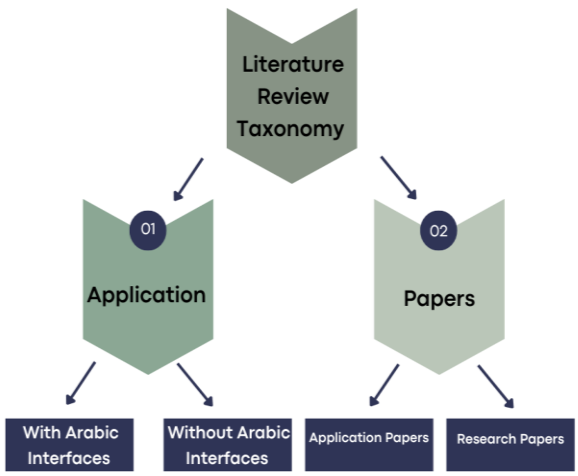


Fig. 1. Literature Review Taxonomy

2.1 Research Papers

In [9] involved a methodology was involved that analyzed current financial industry practices and targeted customers to understand spending patterns with reduced spending. It is helpful for those building systems or improving financial decision-making methods. It tries to solve the problem of poor financial management and planning, lack of knowledge, and inaccurate expense tracking. The authors use knowledge and technology to develop a financial management tool that provides clients with critical insights, financial plans, and monthly income and expense analysis. In [10] solved the problem of how artificial intelligence can assist in improving the allocation of the public budget. The technology that the authors applied is the employment of a hybrid AI approach involving a multilayer perception and multi-objective genetic algorithm to analyze the effects of various public spending categories on inflation, GDP, and income inequality. It uses real-world economic data to predict the outcome of scenarios that result from different budget allocations.

In [11] authors focused on considering the problem of predicting fiscal crises and the limitations of traditional econometric models in accurately forecasting such rare events. The main method proposed is applying machine learning models: random forest, gradient boosted trees, and elastic net, through which the investigator investigates the interactions of the economic, political, and demographic predictors while addressing overfitting through out-of-sample validation and data pooling. In the study of [12] authors discussed applying machine learning techniques to improve financial forecasting and planning. Much focus has gone to supervised learning in predicting financial outcomes based on historical data; unsupervised learning, mainly for pattern recognition; and reinforcement learning, especially for adaptive financial decision-making. It was found that machine learning is good at finding patterns in financial data, but it struggles to explain why those patterns happen, which is important for making decisions.

In [13] authors focused on considering the problem of forecasting stock price movements within the LQ45 financial sector index and the limitations of traditional forecasting models in capturing the highly dynamic and nonlinear nature of financial markets. The main results of this paper show that the LSTM-based approach significantly improves the forecasting accuracy, especially for BBCA and BMRI stocks, which showed the lowest mean relative absolute error (MAPE) values, while other stocks showed reasonable forecasting results.

2.2 Application Papers

In [14] authors tried to solve the problem of people's poor personal financial planning that leads to an inability to meet needs. The authors use existing technology-based methods to create a mobile-based application called Money Empire. Money Empire has many features that distinguish it which are it is a mobile app-based smart assistant for personal finance management. This application automates finance management by extracting transaction details from banking SMS alerts and expense details from invoices. In [15] authors solve the problem of people's inability to manage money properly and not understanding the financial risks that a person may face due to making some wrong decisions. The authors used existing technology-based approaches, such as Google Cloud Vision API and One Signal API, and integrated them into a mobile-based application called Manage on Money. The application received great satisfaction, which helped users manage money.

In [16] authors discussed the problem for people who do not have enough financial knowledge to manage their money and track their daily income. The authors used the iterative development methodology, Android Studio to design the application interfaces, and Firebase for the database. The project was developed with an object-oriented approach. The application manages money and analyzes cash flow effectively. The main results of this paper are an application that manages money, analyzes cash flow effectively, helps users enter and track their spending, and creates financial plans for retirement and others. In [17] authors developed a mobile application for personal finances, offering a set of services for budget tracking, income and expense management, and report generation. The authors used technologies like Java, Gradle 8, and Microsoft SQL to create an Android application, validated through beta testing with real users, including those with visual impairments. As a result, a mobile application was developed that offers a user-friendly interface and useful functionalities. In [18], the author's aimed to create and show a prototype for a mobile application that would manage personal finances, with an emphasis on goal setting, needs organization, and budgeting. The procedure was broken down into two phases: content realization, which realized all the feature lists and specifications from the first phase, and concept formulation, which resulted in an application design with a set of needs and features. Consequently, two outputs were generated from the work: use case and class diagrams from the Content Realization phase and a feature list from the Concept Formulation phase that included Getting Started, Details for Every Function, Goal Setting, Budgeting, Organize Needs, Type of Expense, Indicator for Budgeting, Transaction History, and Reminder.

In [19], the authors developed the Cash Save app as an effective way to manage money, using a prototyping approach with continuous improvements based on user feedback. The app passed 78% of the test cases, demonstrating its overall efficiency, but it faced challenges such as the history tracking feature failing in some cases, requiring additional improvements. In [20] authors developed an Android application for personal finance management, specifically targeted at students. They used the Waterfall Model to design the application, which tracked, categorized, and generated expense reports with customizable daily, weekly, and monthly options. The application effectively met its intended purpose, but the limitations of the Waterfall Model, particularly its lack of adaptability, were noted. Iterative methods were suggested to improve flexibility during the development process. Compared to similar applications, this one emphasized simplicity while integrating AI-driven financial advice to provide personalized recommendations. In [21] identified a challenge was identified in managing personal finances in today's digital world, where using multiple accounts and traditional recording methods can increase financial stress. To address this issue, they introduced MyFinanceAI, an intelligent system that combines deep neural networks, reinforcement learning, and advanced AI technologies. The system features a secure data collection layer, a financial pattern analysis engine, and predictive models for analyzing time series data. It provides personalized recommendations with 40 % more accuracy thanks to its intelligent recommendation engine. The research showed that MyFinanceAI reduced financial stress by 43% and increased monthly savings by 22%. Additionally, 78% of users achieved their financial goals for debt reduction and savings, and expense projection accuracy improved by 30%.

2.3 Applications with Arabic Interfaces

The Wafeer application [22] is an application that helps people make decisions on budget and save money and eliminate wastage and other expenses that are not necessary. A mobile application that can work on Android and iOS. The main techniques and methodology applied to provide several services such as Expense tracking, preparing a personal budget, Expense analysis, and financial alerts, Daily, monthly, and yearly analysis of operations, Chart of spending rate for 6 months and specific periods in the application, make statistics on your categories and purchases through stores, Savings goals are determined by the person.

In the Wise Budget application [23], which is an application that helps people to effectively monitor their spending and optimize savings. Available on both Android and iOS, it provides a lot of features It contains comprehensive budget management, as it requires entering the monthly income to determine a budget, accurately tracking expenses and financial reports by providing illustrative charts and tables, The ability to add debts or loans to the application to facilitate budget management, Payments are entered manually and no pictures or invoices are inserted. In the Wallet app [24], an app designed to help users monitor their income, expenses, and budgets to improve their financial management, it offers features such as financial data categorization, graphical reports, manual data entry, and bank synchronization. The app relies on features that make tracking expenses easy, helping users adopt better financial habits. However, the app requires time-consuming manual entries and does not provide personalized financial guidance. Additionally, its analytical features are considered too simple for experienced users.

The Amwaly application [25] is an application that helps people solve several problems, such as the ability to organize personal money, monitoring spending and encouraging savings available on Android and iOS. The main techniques and methodologies are applied to provide several services. Among the advantages provided by the application are entering income and determining expenses and debts, if any, and then dividing the expenses into monthly requirements, daily and monthly expense keeping, offering notification of payment days and due charges, displaying financial reports to clarify spending habits, and analyzing the budget. In the Masareef application [26], which is an application that manages money through budgeting, account synchronization, and spending tracking, the application integrates with bank accounts, e-wallets, and cryptocurrencies, allowing users to categorize expenses and generate detailed and accurate reports. It also relies on manual data entry to categorize expenses and prepare reports, but some advanced features require a premium subscription to access them. It is available on smartphones only and provides periodic updates to improve performance and increase efficiency. The application has received positive reviews due to its comprehensive features that help users improve their budget and increase their awareness of their spending patterns.

In the money lover application [27], which is an application that helps users to easily keep track of their finances, it is available on both Android and IOS. The application provides several different methods and features, including an easy way to track the user's money, detailed reports, a clear interface, customizable categories, automatic bill reminders, debt management, multi-currency support, a built-in calculator, and more. The Money manager application [28] is an application that solves the problem of the complication of managing the household accounts(budget). Available on both Android and iOS, it provides a lot of features like the ability for the user to enter her/his monthly income and calculate his/her expenses in various branches. The application is very good, and the interfaces are user-friendly. In contrast to “Hassalah”. The money manager application only provides the ability to record his/her purchases in an orderly manner without creating a budget.

2.4 Applications without Arabic Interfaces

In Mobil’s applications [29], it is an application that helps people track expenses and maintain steady savings. It is available both for Android and iOS and provides many features such as adding accounts and credit cards, creating a monthly plan, having a category to organize expenses, adding payment or purchase operations manually, encouraging monthly savings, and providing illustrations of categories. The application received a high rating due to the features it offers. In GoodBudget application [30] which is employing the digital version of envelope budgeting through the GoodBudget application to handle their money. Users can distribute their income across separate expenditure categories through this application without relying on cash payments and access their budgets everywhere because they can synchronize their content between their different devices. Nevertheless, manual financial expense input in the free version is time-intensive for users who are looking for a free application. The application provides an intuitive interface that leads users to control their financial choices. However, the proposed application extends its capabilities by implementing alert notifications when the user is close to exceeding their budget and providing a personalized plan.

In the Moneon application [31], which is an application that helps users manage their personal finances, it is available on Android with iOS. The application provides several different methods and features that serve the user in reaching and achieving goals, including providing personal finance management, unlamented numbers of wallets, full control over categories/subcategories creation, and much more. The application is very good, and it has an iOS 10 interface designed according to Apple's guidelines. In the money flow application [32], which is an application that helps users to keep track of their expenses and income, it is available on Android with iOS. The application provides several different methods and features that serve the user in reaching and achieving goals, including quick and easy adding of transactions, customizable accounts and categories, Synchronization between devices and adding goals and planning the budget, and more. The money manager application [33] is an application that solves the problem of keeping track of users' budgets, finances and keeping the money under control. Available on both Android and iOS, it provides a lot of features like payment reminders, detailed reports, a clear interface, customizable categories, helps the user to make a budget, multi-currency support, and more. The Spendee application [34], which is an application to help users manage their money and track their spending, offers a simple interface that allows for categorizing expenses, generating accurate reports, and setting reminders. The app relies on easy-to-use tools to facilitate personal money management, giving users a clearer view of their spending habits and helping them improve their financial management. However, the app is only available on iOS, and it is noted that it lacks the advanced features needed for professional financial management or business use. In Paymaster application [35], which is an application that helps people manage their financial affairs and maintain their monthly budget, was available on Android and iOS, the application provided several different methods and features that serve the user in reaching and achieving goals, such as computing the monthly installments or splitting the expenditure, they will notify you for payment dates and upcoming bills, preparing the daily and monthly budget, provided accurate analyses and details about spending through graphs and reports, there are some negatives such as manual entry of bills, a difficult interface for the user, the application received a high rating due to features. In the Money application in [36], the application helps the users to keep track of their finances, it is available on Android with iOS. The application provides several different methods and features that serve the user in reaching and achieving goals, including a quick and effortless way to plan your income and expenses, tracking debts and savings, synchronization to track your finances across all your devices, and more. In the Buddy application [37], which is designed to simplify both individual budget management and group finances by providing tools for budget setup, spending monitoring, and shared fund tracking. Despite being easy to use and showing financial transactions, the application doesn't provide alert notifications when the user is close to exceeding the budget, nor offer any entertainment ideas. The proposed application focuses specifically on budgeting duties through personalized financial plans for the user and real-time notification systems that maintain user financial stability.

2.5 Budget Mobile Applications Comparison

Table 1 shows the comparison between the reviewed applications (presented in Sections 2.3 and 2.4) and our proposed application in terms of features and other aspects. To address the gap highlighted in previous works, a comparative analysis was conducted between the proposed application and existing budget management applications as presented in Table 1. The proposed system stands out in terms of functionality, intelligence, and user personalization. Unlike most existing apps that offer basic expense tracking and budgeting, our application integrates advanced AI techniques, specifically an RNN to predict financial risk and provide personalized financial plans. Also, it supports both Arabic and non-Arabic users. Additionally, our system offers unique features such as real-time alert notifications, retirement financial planning, entertainment recommendations based on budget, and a smart wishlist system. Furthermore, our app enables multilateral shared budgets, which is rarely supported elsewhere. Overall, the proposed application provides a more comprehensive, intelligent, and culturally inclusive solution for modern financial management.

3. Methodology

3.1 Design of the Proposed System

The design phase translates requirements into a blueprint for implementation. Key design components include three main parts: the architecture model, data preprocessing, and interface design.

3.1.1 Architecture Model

The architecture based on RNN, specifically implemented a bi- directional LSTM network of two hidden layers, each with 128 units followed by a fully connected output layer with a sofmax activation function for 3 class classification task (low/medium/high risk), the used activation function in hidden layer is RELU, the model was trained using Adam optimize and learning rate is 0.001, batch size is 64, and the number of epochs is equal to 50. It describes how a software system is structured and organized. Architectural design involves decisions about the type of application, system distribution, and architectural styles. Architecture is often documented from multiple perspectives, such as conceptual, logical, process, and development views .The following explains the architecture model of the proposed application, as shown in Fig. 2.

- The system architecture is best described as Client-Server, serverless functions, or a microservice for AI processing.
- Client: A mobile application built using React Native with Expo. This client handles the user interface and user interactions.

- Backend (Database): Cloud Firestore (Firebase) acts as the primary backend, managing data storage, real-time synchronization, and user authentication.
- Backend (AI Processing): A separate component developed in Python handles AI-based tasks like financial risk prediction and personalized recommendations. This could be deployed as cloud functions or a micro service that interacts with Firestore. There are numerous advantages of utilizing Python, where it is far less complicated because its interface is a legacy software written in various languages like C, C++, and others. Also, Python was initially made so that it could be extended using compiled code to increase its efficiency.
- The React Native/Expo app communicates directly with Firestore for most data operations (CRUD). For AI-driven insights, the app might trigger a Python backend service (e.g., via HTTPS requests or cloud function triggers), which processes data from Firestore and potentially writes results back or returns them to the app, it enables real-time data synchronization that ensures data stays up to date between different devices, as it provides safe authentication in addition to allowing efficient tracking of user transactions and budget records.

3.1.2 Data Preprocessing

Preprocessing is an essential step in data analysis, and machine learning is data preprocessing. It requires converting unstructured datasets into formats that are standardized, clean, and appropriate for modeling. Preprocessing guarantees that downstream algorithms can efficiently learn from the data by resolving problems like missing values, inconsistent formats, and incorrect entries. Even the most advanced models are likely to yield false or deceptive results in the absence of strong preprocessing. Imputation of missing values, scaling or normalization of numerical features, categorical variable encoding, and data restructuring for model compatibility are examples of common preprocessing tasks.

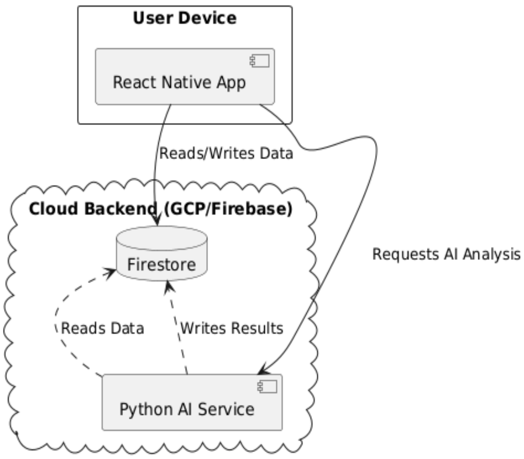


Fig. 2. Architecture Diagram

Table 1. The comparison between the reviewed applications and our proposed application

Categories	Feature / System	Our application	Wafeer	Wise Budget	Mobills	GoodBudget	Buddy: Money Budget	1Money	Moneon	Money Manager	Amwaly	Paymaster	Money Pocket	Daily Pocket	Money Flow	Money Manager	Money lover	wallet app	Monefy	مصاريف	spender
Lang.	Arabic	✓	✓	✗	✗	✗	✗	✗	✗	✗	✓	✗	✓	✓	✗	✓	✓	✓	✗	✓	✗
	Non-Arabic	✓	✗	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✗	✓	✗	✓
Functions	Able to change the icon of recommended expense categories	✓	✓	✓	✓	✗	✓	✓	✗	✗	✓	✓	✗	✗	✓	✓	✓	✓	✓	✗	✓
	Able login with social network	✗	✓	✓	✓	✓	✓	✓	✗	✗	✓	✓	✗	✗	✓	✓	✓	✓	✗	✗	✓
	Able to add goals for saving money	✓	✗	✗	✓	✗	✓	✓	✗	✗	✓	✓	✗	✗	✓	✗	✓	✓	✓	✗	✓
	Additional charge for a specific feature	✗	✗	✓	✓	✓	✓	✓	✓	✗	✓	✓	✓	✓	✓	✗	✓	✓	✓	✗	✓
	Expenses and income tracking	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	Provide a personalized plan for the user	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓
	Alert notifications when is close to exceeding their budget	✓	✓	✗	✗	✗	✓	✓	✓	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	✗	✓
	Recommend entertainment offers based on user's budget	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	✗	✓
	Planned payment	✓	✓	✗	✗	✗	✓	✗	✓	✗	✓	✗	✓	✗	✗	✓	✓	✓	✗	✗	✓
	Budget	✓	✗	✓	✓	✓	✓	✓	✓	✗	✗	✓	✓	✓	✓	✗	✗	✓	✓	✗	✓
	Reinforced filter	✓	✗	✗	✗	✗	✓	✓	✓	✓	✗	✓	✗	✗	✓	✓	✓	✗	✗	✗	✓
	Retirement financial plan	✗	✓	✓	✓	✗	✗	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗
	Bank account sync	✓	✗	✗	✓	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓	✓	✗	✗	✓
	Expenses and income analysis	✓	✓	✗	✗	✓	✗	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	Photo saved in process of logging new expenses	✗	✗	✗	✓	✗	✗	✗	✓	✗	✗	✗	✓	✓	✓	✓	✓	✓	✗	✗	✓
	Export report	✗	✓	✓	✓	✗	✓	✗	✗	✓	✗	✗	✓	✓	✗	✗	✓	✓	✓	✓	✓
	Adding recurring transactions or bills	✓	✓	✗	✗	✓	✓	✗	✗	✗	✗	✗	✗	✗	✗	✓	✓	✓	✓	✗	✓
	wish list	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	✗	✗
	Multilateral shared budgets account	✗	✗	✗	✗	✗	✓	✗	✓	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗	✗	✓

Data Cleaning: A crucial preprocessing step in any data science or machine learning project is data cleaning. Due to a variety of data sources and collection techniques, raw datasets frequently include formatting errors, missing values, duplicate records, inconsistencies, or irrelevant information. By resolving such problems and converting the data into an accurate and structured format, data cleaning aims to improve the dataset's quality, consistency, and dependability. Clean data guarantees that the inputs used in downstream analysis, visualizations, and predictive models are reliable and significant. Ignoring this step may lead to inaccurate insights and subpar model performance. Thus, data cleaning is a fundamental step that directly affects the efficacy and legitimacy of the entire analytical process, rather than merely being a preparatory task.

Standardize Column Names: By changing column names to lowercase and applying underscores (_) as separators, they are standardized to a uniform format. Columns in the code are made more readable and accessible by this method (e.g., Client_Id turns into client_id),

Rename key Linking Columns: The key linking columns, such as id in the user data and client_id in the transaction data, are renamed to a consistent name, user_id. This guarantees a correct connection between the datasets and smooth merging, therefore facilitating accurate table joining.

Parse Date/Time Columns: Date columns are parsed into date-time objects in Python, therefore enabling time-based analysis. Invalid date formats are handled by converting them to NaT (Not a Time), which indicates missing or invalid dates. This makes it possible to increase.

Clean Monetary String Columns: Commas used to group digits, parentheses () that contain negative values, and currency symbols like $ are all removed from monetary columns that contain financial amounts. After that, these columns are transformed into numeric types (float, for example). Unconvertible values are changed to NaN (Not a Number), which denotes inaccurate or missing information.

Risk Label Construction (Low / Medium / High): Because the original Kaggle dataset did not include risk labels, we created them through a clear and replicable scoring method. Each transaction was evaluated using practical financial indicators such as the user's spending relative to income, existing debt pressure, the frequency and timing of transactions, the type of purchase, and whether the spending exceeded the planned budget. These factors helped reflect whether a transaction suggested stable or potentially risky financial behavior. After calculating a combined score for each case, we applied simple thresholds to classify transactions into three levels: low, medium, or high risk, so the labeling process remained transparent, grounded in real financial patterns, and easy to reproduce.

3.1.3 Interface Design

In the budget management proposed application, the interfaces were designed by using Figma, where a focus on providing a smooth and professional user experience. More than 27 interfaces were designed, covering all the features of the app, from registration and login to expense management, financial recommendations, and financial crisis prediction. The design offers an easy-to-use interface that facilitates effective financial control for users.

Log In and Sign-up Interfaces: as shown in Fig. 3. These interfaces represent the basic registration (Sign up), login in the proposed application. The Sign-Up interface includes entering user information to create a new account. The Log In interface allows entering an email and a password.

Profile Page and Edit Profile: The profile interface displays basic user information, with a red "Edit Profile" button. The page also contains several options, such as "Transactions", "Debt/Loan," and language change buttons (Arabic / English), allowing the user to switch between languages easily. As for the edit profile interface, it contains user information fields with a "Save" button to save the changes, as shown in Fig. 4.

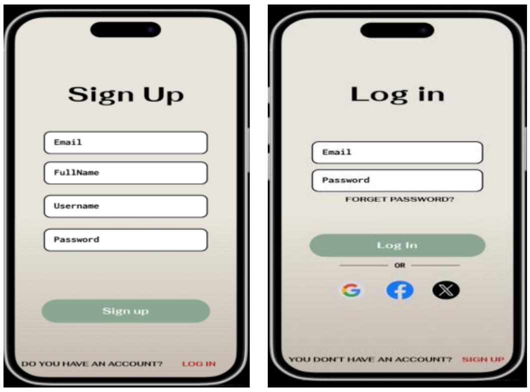


Fig. 3. Sign UP and Log in Interfaces

Fig. 4. Profile Page Interface

Report Interfaces: These interfaces represent the management of financial accounts and monthly reports, which the user can access through the menu bar at the bottom of the page user can add new account, as shown in Fig. 5.

Budget Interfaces: These interfaces represent the budget management pages (My Budget) in the proposed application, where the user can easily control their money and track their spending, and add a new budget, as shown in Fig. 6.

3.2 Implementation phase

The implementation phase involves the practical development of the proposed application using selected tools, technologies, and methodologies to achieve the defined functional and non-functional requirements. Below is a detailed breakdown of the implementation process. For programming Language & Tools, the following tools and technologies are planned for the development of the proposed application:

Frontend Development:
React Native: Framework for building cross-platform mobile applications using JavaScript/TypeScript. Expo Toolset and platform built around React Native to streamline development, building, and deployment.

Backend Development :
Python is selected for implementing the AI-driven components of the application, including financial crisis prediction and personalized recommendations [48]. Rich Ecosystem: Libraries like TensorFlow/Keras (for building Recurrent Neural Networks) and scikit-learn streamline proposed model development.

3.3 The proposed system

The transaction type and category were used as the input features, while the day of the month, time between transactions, and transaction sequence order were normalized and fed into the RNN to capture the sequence of financial activities. The proposed application combines back-end development using a Python Flask API with front-end development using a React Native mobile application. To provide real-time spending risk predictions, the API combines a pre-trained machine learning model, encapsulates business logic, and coordinates data persistence in Firestore.

3.3.1 Backend Development

The backend architecture of the proposed application was developed using Python and the Flask framework. It provides a connection point for the user interface, the database, and the AI model. The backend acquires application transaction data to perform required feature engineering before providing input to the trained model to make financial risk predictions. The results are then stored in a Firestore database and returned to the application interface for display to the user. This architecture provides an efficient and secure API that enables the system to operate smoothly and provide accurate predictions in real time.

Firebase Firestore: In the application, Firebase Firestore is used as a cloud database where all the user data is stored in an organized, flexible, and automated way. The data in Firestore is split up into multiple main collections, with each collection containing documents that represent various data entities.

Users: This collection includes basic user information, including name, email, age, and annual income. This data is used to build a personal budget and analyze risks. This data is entered and saved in the Fire Store when the user registers from the sign-up page.

Transactions: This collection stores all the financial transactions that the user has made, either expenses or income. Each document contains the type, amount, date, associated account, and category. New transactions can be added from the Add page.

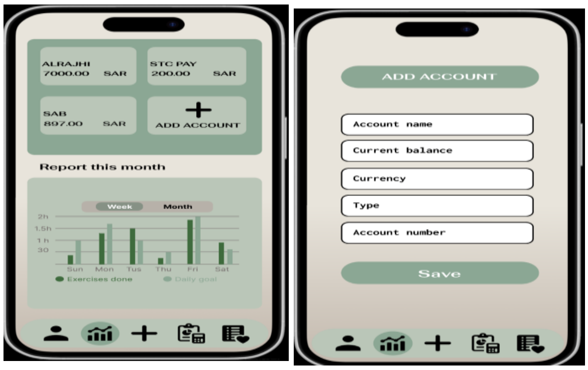


Fig. 5. Report interface and Add Account Interface

Fig. 6. My Budget Interface and add a new Budget interface

Budget: This collection stores the budget amount for each category, such as food, entertainment, and transportation etc. The data is later used by comparing the actual spending against the amount specified.

Loan and Debt: These two collections contain debts and loans, and store the other party's name, amount, due date, and payment status. A new loan or debt is added from the Add Loan and Add Debit pages, and this data is used to track due and expected payments.

Account: This collection represents all the accounts owned by the user and contains the account name and account amount. A new account can be added from the Add Account page.

Recurring Transactions: This collection is used to save recurring transactions such as subscriptions or monthly payments, for easy addition or late.

Wishlist: This collection contains the user's wish list. Every item has the name, brand, price, priority, and fulfilled/not fulfilled completion status. New items can be added from the add item page.

Financial Risk Prediction: This collection records the results of the smart analysis of each transaction based on the artificial intelligence model, and contains the risk level (low, medium, high) and the probability of each.

Firebase Authentication: Helped to offer a secure and simple sign-in and sign-out system to users. It signs in using email & password and saves login details, including login time and last login time.

3.3.2 Model Integration

After the neural network model was trained and saved, it was deployed in the application using a backend framework built with Flask. When a user adds a transaction in the app, the backend receives this data and saves it to the cloud database (Firestore). It then collects all the users' recent transactions for the current month and tries to find a monthly budget. If no budget has been set before, the system automatically calculates one based on the user's income. This data is then passed to a special feature-engineering module that turns it into numbers the model can understand, just like the ones used during training. The backend loads the trained model, the scaler (which keeps number ranges consistent), and a list of the features to expect. The features are sent to the model, which returns the predicted financial risk level (low, medium, or high). If the transaction is an income, the risk is always set to "Low" automatically. The feedback from the AI model is displayed in the app immediately and gets saved to the FINANCIAL_RISK_PREDICTION. For the integration to start, the required Python libraries and packages were installed from the requirements text file. This file contains all that is needed to make the AI model work. For example, Flask is used to build the API, which will connect the app to the model, while Flask-CORS enables safe communication of different domains (app to backend in our case). TensorFlow ensures that there are necessary tools to load and run the trained neural network. Scikit-learn is used for jobs such as data pre-processing and model support utility. Pandas and NumPy help deal with data and work on it.

3.3.3 Fronted Development

The front end of the proposed application was developed using React Native with Expo, enabling cross-platform functionality for both Android and iOS using a single codebase. The primary objective was to create an interface that is clean, intuitive, bilingual (Arabic and English), and responsive to real-time financial data from the backend AI system.

React Native is an open-source framework developed by Meta (formerly Facebook) that allows developers to build mobile applications using JavaScript and React. Unlike traditional mobile app development approaches (which use Java/Kotlin for Android and Swift/Objective-C for iOS), React Native enables cross-platform development—write once, run on both platforms.

Main Interfaces: The proposed application contains several core interfaces that will assist the user in managing his / her finances effectively. These screens cover such main functions as adding transactions, managing budgets, as well as setting financial goals.

Error Handling: The proposed application includes relevant error handling to guide users better while navigating the app. Validation on mandatory fields is carried out, and alerts come with pertinent information for users when selected actions fail.

Confirmation & Success Feedback: The proposed application provides users with simple notifications once essential tasks are executed, including saving transactions or adding accounts. These kinds of confirmations are visible to users in the form of alerts/pop-ups to reassure them.

3.4 Model Performance

The performance of the proposed application is measured based on Precision, Recall, F1-score, and accuracy.

- Precision was calculated as the number of true positives divided by the sum of true positives and false positives:
- $\text{Precision} = \frac{\text{TruePositives}}{\text{TruePositives}+\text{FalsePositives}}$

- Recall was computed by dividing true positives by the sum of true positives and false negatives:

$$\text{Recall} = \frac{\text{TruePositives}}{\text{TruePositives} + \text{FalseNegatives}}$$

- F1-score, which balances precision and recall, was calculated as:

$$\text{F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

- The accuracy score represents the proportion of correct predictions out of all predictions:

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}}$$

These formulas helped ensure the evaluation was both quantitative and interpretable, providing meaningful insights into model behavior across risk levels.

- Stability / Error Rate: How frequently do users encounter bugs or crashes? The aim is to minimize technical errors.
- Data Security: (Implicit Requirement) Is user data stored and handled securely within Firestore and during API interactions?

4. Experimental Results

4.1 Dataset Description

The main data source used for this project was the Transactions Fraud Datasets published on Kaggle [38]. This dataset contains two tables: the transaction table and the user's table. The transactions table, with 1,048,576 real financial transactions that contain the transaction details like date, amount, client_id, etc. The users table contains personal and financial information about 2,000 users, like age, yearly income and total debt. The dataset was divided into 70% training, 15% validation, and 15% testing sets. Although the original purpose of this dataset was to detect fraud, it was used in this paper to design an intelligent model capable of assessing the financial risk level of each transaction based on user behavior. The dataset includes transaction details about amounts, transaction type, and payment methods, along with information about users such as age, gender, income status, and debt data. The selected dataset was useful for building predictive models because it contained extensive, analyzable features that reflected actual financial behavior patterns. Testing software is an essential task in software engineering that guarantees good quality and reliable applications. Functional Testing: Functional testing was used to ensure all the features worked based on the requirements specifications. User registration, login attempts, adding accounts, adding a budget, risk alert, and notification were verified. All core functionality was evaluated, including: The following table shows examples of test cases used during functional testing.

4.2 Prediction Model Results

To assess the model's ability to predict spending risk levels, a separate set of unseen financial transactions was used to replicate a real-world application. The results indicate that the model performs with high reliability, particularly in detecting low and high-risk behaviors, giving users greater confidence in managing their finances. Its sensitivity to medium-risk patterns is solid, though slightly more variable, reflecting the nuanced nature of such behaviors. Table 2 presents a summary of core performance metrics, including precision, recall, and F1-score. As illustrated in Table 3, the confusion matrix highlights how closely the model's predictions aligned with actual outcomes. The majority of predictions in each category, especially "Low" and "High," were correct, while slight overlaps occurred within the "Medium" category. Meanwhile, these results confirm that the model not only offers reliable classifications but also serves as a strong foundation for continuous improvement and refinement in risk assessment.

4.3 Application Testing Case

Functional testing was used to ensure all the features worked based on the requirements specifications. User registration, login attempts, adding accounts, adding a budget, risk alert, and notification were verified. All core functionality was evaluated, including: The following Table 4 shows examples of test cases used during functional testing. For the proposed application, all the results were good and reflect the app's success. The app was tested with test cases, including the most important features, such as registering a new account, adding a new transaction, adding a budget, adding accounts, and more. The application passed all these tests without errors, which indicates the stability of the application and its ease of use.

Table 2. Classification Report

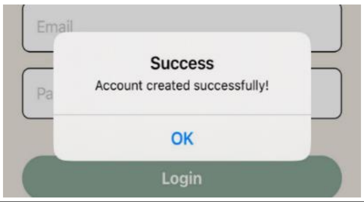
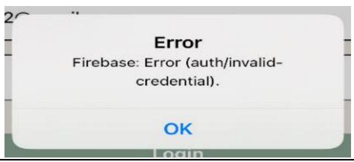
Risk Level	Precision	Recall	F1-Score	Support
Low	0.97	0.97	0.97	588,091
Medium	0.97	0.97	0.97	619,864
High	0.98	0.98	0.98	723,206
Macro Avg.	0.97	0.97	0.97	1,931,161
Weighted Avg.	0.97	0.97	0.97	1,931,161

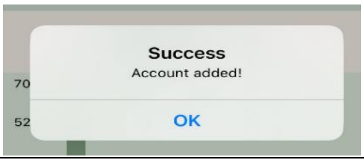
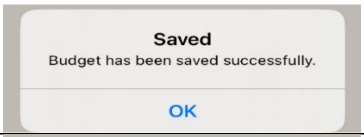
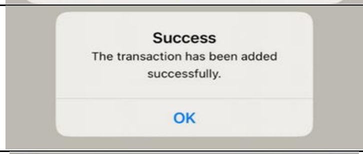
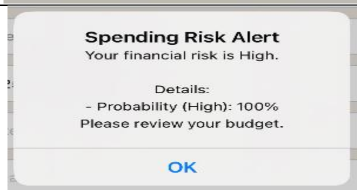
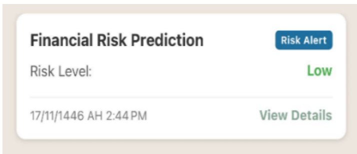
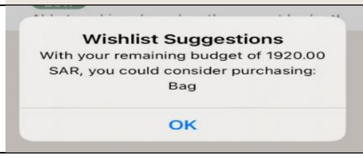
Table 3. Confusion Matrix

	Predicted Low	Predicted Medium	Predicted High
Actual Low	569,272	7,300	11,519
Actual Medium	16,331	603,108	425
Actual High	2,112	11,596	709,498

All of this confirms that the application is a useful tool in the daily lives of users. The prediction model's outcomes show a real technical victory, but they also represent a significant human step toward empowering consumers to make better financial decisions. When evaluated on new data, the model attained an overall accuracy of 97.45%, demonstrating its robustness and high potential for practical use. Precision, recall, and F1 measures across all risk categories (low, medium, and high) outperformed 0.97, confirming the model's ability to accurately detect expenditure patterns.

Table 4. Functional Testin

Test case ID	Scenario	Input/Test Steps	Expected Output	Status	Actual Output
TC-1	New Registration	1. Open the app 2. Press "sign up." 3. Enter all information 4. Press "Sign up."	User Register successfully stores all the user information in the database, and the user is redirected to the login interface.	Pass	
TC-2	Log in	1. Open the app 2. Press "Log in." 3. Enter Email and password 4. Press "Login"	The user logs in, and the user is redirected to the Home(profile) interface.	Pass	As expected,
TC-3	Failed login	1. Enter an incorrect Email and password 2. Press "Login"	An error message pops up on the screen.	Pass	

TC-4	Add a new account	1. Go to the report interface 2. Press "add account." 3. Enter all the required information 4. Press "save"	The account is created message and saves the info into the database.	Pass	
TC-5	Add new budget	1. Go to the "My Budget "interface 2. Enter the month and assign a budget for each category 3. Press "save"	The Budget is created message and saves the info into the database.	Pass	
TC-6	Add new Transaction	1. Go to the "Add" interface 2. Enter all the required information 3. Press "save"	The Transaction is created message and the info is saved into the database.	Pass	
TC-7	Financial alert (high)	1. Go to the "Add" interface 2. Enter a high amount 3. Enter all the required information 4. Press "save"	An alert will pop up.	Pass	
TC-8	Financial alert (medium)	1. Go to the "Add" interface 2. Enter a reasonable amount 3. Enter all the required information 4. Press "save"	A medium alert will pop up.	Pass	
TC-9	Financial alert (Low)	1. Go to the "Add" interface 2. Enter a small amount 3. Enter all the required information 4. Press "save"	No pop-ups, but it will be added to the notification interface.	Pass	
Pass	Suggestions of what to buy from the wish list items	1. Go to the "Wishlist" interface 2. Press the money-shaped button at the top of the page	The suggestion will pop up.	Pass	

5. Conclusion, recommendations and future work

The proposed application aimed to develop a bilingual smart budget management mobile application designed to enhance personal financial planning through artificial intelligence and address a critical gap in the market. It supports Arabic and English users by offering an accessible interface and advanced features such as expense tracking, debt management, budget monitoring, and AI-powered financial risk prediction. The development of the lifecycle, covering requirements gathering, system design, implementation, and testing, the application demonstrated both functional completeness and high usability. The proposed application achieved 97.45% accuracy in classifying financial transactions by risk level (low, medium, high), enabling real-time alerts that empower users to make informed decisions and avoid financial stress.

Although the proposed application achieved its intended goals, there remain valuable directions for future improvement and expansion.

New Features: Future versions of the app could include:

- Integration with online payment systems to automate financial tracking and improve convenience.
- Entertainment recommendations that fit within the user's available budget.
- Joint/multilateral budget sharing, allowing families or groups to collaboratively manage shared expenses.

1. Scalability: Expanding the proposed application to support multiple currencies and regional tax structures could enable its adoption beyond the initial target audience. Moreover, developing a web-based companion interface could improve accessibility.
2. Technological Enhancements:

- Bank account synchronization would allow real-time financial data import, reducing manual entry.
- Blockchain integration could improve transaction transparency, data integrity, and traceability in budget management processes, thereby enhancing the overall accountability and security of financial operations.
- Advanced AI techniques, such as generative AI or reinforcement learning, can improve personalization and forecasting accuracy.

3. Deployment: Publishing the proposed application on the App Store and Google Play is a necessary next step. This will require addressing platform-specific requirements,
4. Ensuring security compliance and setting up user support systems.
5. Extended Applications: The core architecture of the proposed application could be adapted for broader use cases such as small business budgeting, community saving circles, or educational tools promoting financial literacy among students.

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